

Python meets AI

1-Hour Lab · Semester 1 · Python + AI Basics

5
Exercises

~1 Hour
Duration

Semester 1
Level

Topics covered: lists, functions, strings, if/else · AI concepts: datasets, labels, data cleaning, classifiers, prediction

Exercise 1 ◆ Python

My First Dataset

Tags: lists · indexing

AI AI concept — Dataset: In AI, a dataset is a collection of data used to train or test a model. In Python, we can represent small datasets as lists.

Create a list of 5 student names. Print the first name, the last name, and the total count of students.

```
students = ["Ali", "Sara", "Ahmed", "Zara", "Bilal"]

# print the first student
# print the last student
# print total number of students
```

Hint: Use index [0] for first, [-1] for last, and len(students) for the count.

Exercise 2 ● AI Concept

Label Your Data

Tags: lists · if/else · functions

AI AI concept — Labels: In AI, labels are the answers we want the model to learn. For example, emails are labelled "spam" or "not spam". Here we label temperatures.

Given a list of temperatures, write a function that labels each one as "hot", "warm", or "cold". Hot = above 35, Warm = 20–35, Cold = below 20.

```
temperatures = [42, 18, 27, 35, 10, 30]

def label_temp(temp):
    # return "hot", "warm", or "cold"
    pass

for t in temperatures:
```

```
print(t, "->", label temp(t))
```

Hint: Use if/elif/else inside the function. Replace "pass" with your conditions and return statements.

Exercise 3 ◆ Python

Clean the Data

Tags: strings · strip() · lower()

AI AI concept — Data Cleaning: Before AI models are trained, data must be cleaned — removing extra spaces, fixing capitalisation, and standardising format. Dirty data leads to bad models.

You have a messy list of city names with extra spaces and mixed case. Clean each one so they are all lowercase with no extra spaces.

```
cities = [" Karachi ", "LAHORE", " islamabad", "Peshawar ", " QUETTA "]

def clean(city):
    # remove spaces and make lowercase
    pass

clean_cities = [clean(c) for c in cities]
print(clean_cities)
```

Hint: Use .strip() to remove whitespace and .lower() to make lowercase. Chain them: city.strip().lower()

Exercise 4 ● AI Concept

Simple Classifier

Tags: functions · lists · if/else

AI AI concept — Classifier: A classifier is an AI model that sorts input into categories. Spam filters, disease detectors, and image recognisers are all classifiers. Here you build a tiny one!

Build a spam word classifier. If a message contains words like "free", "win", or "prize", classify it as "SPAM", otherwise "NOT SPAM".

```
spam_words = ["free", "win", "prize", "click", "urgent"]

def classify(message):
    message = message.lower()
    # check if any spam_word is in the message
    # return "SPAM" or "NOT SPAM"
    pass
```

```
print(classify("You have won a free prize!"))
print(classify("Can we meet tomorrow?"))
print(classify("Urgent: click here now"))
```

Hint: Loop through `spam_words`. Use the `"in"` keyword: if word in message. Return `"SPAM"` as soon as you find one.

Exercise 5 • AI Concept

Predict the Score

Tags: [functions](#) · [lists](#) · [loops](#)

AI AI concept — Prediction: Prediction is the core job of AI — given some input, estimate an output. The simplest prediction is an average. Real AI models do something much more complex, but the idea is the same.

Write a function that predicts a student's final grade: A (90+), B (75-89), C (60-74), or F (below 60) based on the average of their scores.

```
def average(scores):
    # calculate and return the average
    pass

def predict_grade(scores):
    avg = average(scores)
    # return "A", "B", "C", or "F"
    pass

ali_scores = [88, 92, 85, 90]
sara_scores = [70, 65, 72, 68]
ahmed_scores = [50, 55, 48, 60]

print("Ali:", predict_grade(ali_scores))
print("Sara:", predict_grade(sara_scores))
print("Ahmed:", predict_grade(ahmed_scores))
```

Hint: For average: use `sum(scores) / len(scores)`. Then use `if/elif/else` on the result to return the grade letter.

Complete each exercise in Python. Use the hint only after you have tried on your own. Mark each exercise done when finished.